

Applied Statistics & Business Analytics

End-Course Summative Assignment

Topic:Top 50 interview questions

Student Name: Sumit Mishra

Course: Applies Statistics & Business Analytics

Instructor: Krishnendu Mandal

Institution: ALMABETTER

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1. Introduction

This document presents professional solutions to fifty selected interview questions from the Applied Statistics and Business Analytics curriculum.

Each answer is concise, technically accurate, and demonstrates conceptual understanding expected in academic and industry interviews.

The final section provides five questions explained in a recorded video for deeper conceptual clarity.

2. Interview Questions and Answers

1. What is a vector in mathematics?

A vector is a quantity that has both magnitude and direction. It is represented as an ordered set of numbers that indicate a position or movement in space.

2. How is a vector different from a scalar?

A scalar has only magnitude (e.g., 5 kg), while a vector has both magnitude and direction (e.g., 10 m/s north).

3. What are the different operations that can be performed on vectors?

Addition, subtraction, scalar multiplication, dot product, and cross product.

4. How can vectors be multiplied by a scalar?

Multiply each component by the scalar:

$$2 \times (3, 4) = (6, 8).$$

5. What is the magnitude of a vector?

$|v| = \sqrt{x^2 + y^2 + z^2}$. It represents vector length.

6. What is the difference between a square and a rectangular matrix?

Square matrix $\rightarrow$ rows = columns.

Rectangular matrix $\rightarrow$ rows $\neq$ columns.

7. What is a basis in linear algebra?

A set of linearly independent vectors that span a vector space. Every vector is a combination of basis vectors.

8. What is an eigenvector?

A vector that changes only in magnitude (not direction) when a matrix transformation is applied.

9. What is a gradient in machine learning?

A vector of partial derivatives indicating the direction of steepest increase of a function. Used in gradient descent optimization.

10. What is backpropagation?

Algorithm that updates weights in neural networks by propagating errors backward and computing gradients.

11. What is probability theory?

Mathematical framework for quantifying uncertainty and likelihood of events.

12. What is conditional probability?

$P(A | B) = P(A \cap B) / P(B)$; the chance that A occurs given B has occurred.

13. What is Bayes' Theorem?

$P(A | B) = [P(B | A) \times P(A)] / P(B)$; updates beliefs with new evidence.

14. What is a random variable?

A variable whose values depend on outcomes of a random experiment.

15. What is the Central Limit Theorem?

For large samples, the sampling distribution of the mean is approximately normal, regardless of population distribution.

16. What is the Law of Large Numbers?

As sample size increases, sample mean converges to the true population mean.

17. What are measures of central tendency?

Mean, median, and mode summarize the center of a dataset.

18. What is an outlier and how is it detected?

A value far from others; use Z-score ($|Z| > 3$) or IQR rule ($1.5 \times \text{IQR}$).

19. What is sampling and why important?

Selecting a subset to represent a population — saves time and resources while allowing valid inference.

20. What are sampling methods?

Random, stratified, systematic, cluster, and convenience sampling.

21. What is parameter estimation?

Using sample statistics to estimate unknown population parameters.

22. What is hypothesis testing?

Statistical procedure to decide whether sample data supports a claim about a population.

23. What is a p-value?

Probability of observing data as extreme as the sample if the null hypothesis is true.

24. What are Type I and Type II errors?

Type I (α): Reject true H_0 . Type II (β): Fail to reject false H_0 .

25. What is a confidence interval?

Range of values likely to contain the true population parameter at a given confidence level (e.g., 95%).

26. What does confidence level represent?

Proportion of times CIs from repeated samples would contain the true parameter.

27. What is correlation vs causation?

Correlation = association; Causation = cause–effect relationship. Correlation alone cannot prove causality.

28. What is experimental design?

Planning an experiment to obtain unbiased, reliable data for causal inference.

29. What are key elements of experiment design?

Control group, randomization, replication, blinding, and adequate sample size.

30. How does sample size affect experiments?

Larger samples reduce variance and increase power.

31. How to mitigate bias?

Randomization, double-blind designs, representative sampling, standardized procedures.

32. What is a Chi-Square test?

Evaluates association between categorical variables or fit of observed to expected frequencies.

33. What is a t-test?

Compares means of two groups to determine statistical difference.

34. What is ANOVA?

Compares means of three or more groups to detect significant differences.

35. What is Bessel's correction?

Use $n-1$ instead of n in sample variance to reduce bias when estimating population variance.

36. What is kurtosis?

Degree of tailedness of a distribution; high = heavy tails, low = light tails.

37. What is skewness?

Measure of asymmetry. Right-skew = long right tail; left-skew = long left tail.

38. What is the difference between descriptive and inferential statistics?

Descriptive → summarize data; Inferential → draw conclusions about population from samples.

39. What is sampling error?

Gap between sample statistic and true population parameter; reduced by larger random samples.

40. What is a one-tailed vs two-tailed test?

One-tailed checks effect in one direction; two-tailed tests both directions.

41. What is statistical power?

Probability of correctly rejecting a false null hypothesis ($1 - \beta$).

42. What is the Empirical Rule?

For normal data: 68% within 1 SD, 95% within 2 SDs, 99.7% within 3 SDs of mean.

43. How does sample size affect CI width?

Larger sample → smaller standard error → narrower confidence interval.

44. What is a joint probability distribution?

Probability distribution showing likelihood of two variables occurring together.

45. What is covariance?

Indicates whether two variables move together (positive) or opposite (negative).

46. What is correlation coefficient?

Standardized covariance from -1 to $+1$ indicating strength and direction of linear relation.

47. What is a random variable?

Maps random events to numerical outcomes (discrete or continuous).

48. What is the difference between discrete and continuous variables?

Discrete $\rightarrow$ countable values; Continuous $\rightarrow$ infinitely many values in a range.

49. How is hypothesis testing used in A/B testing?

Compare two variants (A vs B) to see if performance difference is statistically significant.
 H_0 : no difference; H_1 : difference exists.

50. What is margin of error in a CI?

Maximum expected difference between sample statistic and true population parameter.

1. Central Limit Theorem (CLT)

Concept:

The CLT says that when we take large random samples from any population, the **mean of those samples** will follow an **approximately normal (bell-shaped) distribution**, no matter how the original data looks.

Formula:

$$\bar{X} \sim N(\mu, \sigma^2/n) \quad \bar{X} \sim N(\mu, \sigma^2/n)$$

Key idea:

As sample size increases:

- The sample mean becomes more stable.
- The shape of the sampling distribution becomes more normal.

Example:

If you take the average income of 50 people again and again, those averages will form a normal curve even if individual incomes are skewed.

Why it matters:

It allows us to use normal-based confidence intervals and hypothesis tests even for non-normal populations.

2. Hypothesis Testing

Concept:

It's a structured method to decide whether our data provides enough evidence to support or reject a claim about a population.

Steps:

1. Set hypotheses:

H_0 (no effect) and H_1 (there is an effect).

2. **Select significance level (α):** usually 0.05.

3. **Compute test statistic & p-value.**

4. **Decision:**

- If $p < \alpha \rightarrow$ reject H_0 .
- If $p \geq \alpha \rightarrow$ fail to reject H_0 .

Example:

To test if average delivery time < 30 mins:

$H_0: \mu = 30$, $H_1: \mu < 30$.

If $p = 0.03$, we reject $H_0 \rightarrow$ evidence that average time < 30 mins.

Why it matters:

It quantifies how confident we are that an observed difference isn't just due to random chance.

3. Confidence Intervals (CIs)

Concept:

A confidence interval gives a **range of plausible values** for an unknown population parameter.

Formula (for a mean):

$$CI = \bar{X} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \quad \text{CI} = \bar{X} \pm z_{\alpha/2} (n\sigma)$$

If σ unknown, use t instead of z .

Interpretation:

A 95 % CI means if we repeated the study many times, about 95 % of those intervals would contain the true mean.

Factors that affect width:

- Larger $n \rightarrow$ narrower CI.
- Higher confidence level $\rightarrow$ wider CI.
- More variability $\rightarrow$ wider CI.

Why it matters:

Shows both **estimate and uncertainty**, unlike a single point estimate.

4. A/B Testing

Concept:

A real-world application of hypothesis testing where two versions (A = control, B = variant) are compared to see which performs better on a chosen metric.

Steps:

1. Define the success metric (e.g., conversion rate).
2. Randomly divide users into Group A and Group B.
3. Collect data and compare results.
4. Use a statistical test (like a z-test) to check significance.

Example:

Version A conversion = 8 %, Version B = 9 %, $p = 0.04 \rightarrow$ B performs significantly better.

Why it matters:

A/B testing helps make **data-driven decisions** in marketing, product design, and UX improvements.

5. Correlation vs Causation

Concept:

- **Correlation** means two variables move together.
- **Causation** means one variable directly affects the other.
A strong correlation **does not prove** causation.

Example:

Ice-cream sales and drowning cases rise together—but both are caused by higher temperatures, not by each other.

How to test causation:

- Conduct controlled experiments.
- Use causal study designs like **A/B tests, time-series (before-after), or control groups**.

Why it matters:

Distinguishing correlation from causation prevents wrong business conclusions.

Conclusion ~

This project reinforced my understanding of core concepts in Applied Statistics and Business Analytics—from sampling and inference to experimental design and causal

reasoning. Developing detailed explanations and visual aids enhanced my analytical communication skills, critical for real-world data-driven decision-making.